

*****Default*****

Testing Google Cloud Free
Generating test-top.uf2 for the pico-ice
RP2040 & iCE40UP5K FPGA
leddigits
PipelineC blink-ex
PipelineC blink-ex-uart-echo
10/22/25

*****Default*****

First the Google Cloud VPS is created. This takes several steps found in
~/samba_info25/oct_eom/google-cloud.doc. Once the VPS is created
<https://console.cloud.google.com/compute/instances?project=liquid-engine-468502-s6>

The OSS_CAD_SUITE oss-cad-suite-linux-x64-20251020.tgz provides Yosys Yosys Open
SYnthesis Suite and iCE40-Nextpnr -- a portable FPGA place and route tool

```
vidal_develone@mygoogle-cloud1:~$ uname -a
Linux mygoogle-cloud1 6.1.0-40-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.153-1
(2025-09-20) x86_64 GNU/Linux
vidal_develone@mygoogle-cloud1:~$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
udev            1997928      0 1997928   0% /dev
tmpfs           401892    2000 399892   1% /run
/dev/sda1       82317152 19790164 59057028 26% /
tmpfs           2009444      0 2009444   0% /dev/shm
tmpfs           5120        0   5120   0% /run/lock
/dev/sda15      126678    11840 114838  10% /boot/efi
tmpfs           401888      0 401888   0% /run/user/1000

processor       : 0
vendor_id      : GenuineIntel
cpu family     : 6
model          : 79
model name     : Intel(R) Xeon(R) CPU @ 2.20GHz
stepping       : 0
microcode      : 0xffffffff
cpu MHz        : 2199.998
cache size     : 56320 KB
physical id    : 0
siblings       : 2
core id        : 0
cpu cores      : 1
apicid         : 0
initial apicid : 0
fpu            : yes
fpu_exception  : yes
cpuid level    : 13
wp             : yes
flags           : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush
mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology
nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe
```

popcnt aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch invpcid_single pti ssbd
ibrs ibpb stibp fsgsbase tsc_adjust bmi1 hle avx2 smep bmi2 erms invpcid rtm rdseed adx smap
xsaveopt arat md_clear arch_capabilities
bugs : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs taa
mmio_stale_data retbleed bhi its
bogomips : 4399.99
clflush size : 64
cache_alignment : 64
address sizes : 46 bits physical, 48 bits virtual
power management:

processor : 1
vendor_id : GenuineIntel
cpu family : 6

model name : Intel(R) Xeon(R) CPU @ 2.20GHz
stepping : 0
microcode : 0xffffffff
cpu MHz : 2199.998
cache size : 56320 KB
physical id : 0
siblings : 2
core id : 0
cpu cores : 1
apicid : 1
initial apicid : 1
fpu : yes
fpu_exception : yes
cpuid level : 13
wp : yes
flags : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush
mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology
nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe
popcnt aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch invpcid_single pti ssbd
ibrs ibpb stibp fsgsbase tsc_adjust bmi1 hle avx2 smep bmi2 erms invpcid rtm rdseed adx smap
xsaveopt arat md_clear arch_capabilities
bugs : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs taa
mmio_stale_data retbleed bhi its
bogomips : 4399.99
clflush size : 64
cache_alignment : 64
address sizes : 46 bits physical, 48 bits virtual
power management:

Additional package are added with the script extra64.sh

```
#!/bin/bash
sudo apt-get update
sudo apt-get install octave vlc kicad hexchat \
gawk gimp gtwave iverilog clang \
samba samba-common-bin smbclient cifs-utils \
libftdi-dev libmpc-dev libmpfr-dev squashfs-tools bc mercurial \
```

```

libbison-dev libgmp-dev libelf-dev ncurses-dev \
default-jdk libreadline-dev xdot graphviz minicom xpdf lrzsz \
libusb-1.0.0 exuberant-ctags gparted ffmpeg \
mesa-utils libboost-all-dev freeglut3-dev \
python3-dev libgtk2.0-dev libcairo2-dev libpango1.0-dev \
libgdk-pixbuf2.0-dev libatk1.0-dev libghc-x11-dev binutils-arm-none-eabi \
gcc-arm-none-eabi diffuse gitk tcl-dev telnet tftp hexedit cmake flex bison \
gtkmm-3.0 gperf build-essential nfs-kernel-server \
libeigen3-dev libffi-dev libssl-dev python3-pip libtiff-dev ppp \
gdb-multiarch firefox-esr ninja-build ecryptfs-utils \
liboctave-dev autoconf automake libreoffice indent iperf \
wireshark nmap dnsmasq mosquitto mosquitto-clients fritzing doxygen espeak \
libssl-dev zlib1g-dev libbz2-dev libreadline-dev libsqlite3-dev llvm \
libncurses5-dev libncursesw5-dev xz-utils tk-dev libgdbm-dev lzma \
lzma-dev tcl-dev libxml2-dev libxmlsec1-dev libffi-dev liblzma-dev \
wget curl make build-essential openssl libhdf5-dev libc-ares-dev \
libeigen3-dev gcc gfortran libgfortran5 libatlas3-base libatlas-base-dev \
libopenblas-dev libblas-dev liblapack-dev cython3 \
libatlas-base-dev openmpi-bin libopenmpi-dev python3-dev \
libssl-dev zlib1g-dev libbz2-dev libreadline-dev libsqlite3-dev \
llvm libncurses5-dev libncursesw5-dev xz-utils tk-dev libgdbm-dev \
lzma lzma-dev tcl-dev libxml2-dev libxmlsec1-dev libffi-dev \
liblzma-dev wget curl make build-essential openssl \
libssl-dev zlib1g-dev libbz2-dev libreadline-dev libsqlite3-dev \
llvm libncurses5-dev libncursesw5-dev xz-utils tk-dev libgdbm-dev \
lzma lzma-dev tcl-dev libxml2-dev libxmlsec1-dev libffi-dev \
liblzma-dev wget curl make build-essential openssl \
libusb-dev libreadline-dev libglib2.0-dev libudev-dev libdbus-1-dev \
libical-dev libmosquitto-dev hostapd traceroute help2man gforth \
lld pkg-config python3 libboost-system-dev libboost-python-dev \
libboost-filesystem-dev clang-format mosh

```

```

vidal_develone@mygoogle-cloud1:~/PipelineC$ less .git/config

```

```

[core]
    repositoryformatversion = 0
    filemode = true
    bare = false
    logallrefupdates = true
[remote "origin"]
    url = https://github.com/develone/PipelineC.git
    fetch = +refs/heads/*:refs/remotes/origin/*
[branch "test-dev-050125"]
    remote = origin
    merge = refs/heads/test-dev-050125
[branch "test-dev"]
    remote = origin
    merge = refs/heads/test-dev

```

```

vidal_develone@mygoogle-cloud1:~/PipelineC$ git branch -a
* test-dev
test-dev-050125

```

```
remotes/origin/HEAD -> origin/master
remotes/origin/master
remotes/origin/test-dev
remotes/origin/test-dev-050125
remotes/origin/test-dev-060125
remotes/origin/test-dev-091125
```

```
vidal_develone@mygoogle-cloud1:~/PipelineC$ cd ~/pico-ice/
```

```
vidal_develone@mygoogle-cloud1:~/pico-ice$ less .git/config
```

```
[core]
    repositoryformatversion = 0
    filemode = true
    bare = false
    logallrefupdates = true
[remote "origin"]
    url = https://github.com/develone/pico-ice.git
    fetch = +refs/heads/*:refs/remotes/origin/*
[branch "test-dev-100424"]
    remote = origin
    merge = refs/heads/test-dev-100424
```

The screenshot displays the Google Cloud Platform console interface. The main heading is "VM instances - Compute Engine". Below this, there's a search bar and a "Create instance" button. The left sidebar shows a navigation menu with categories like "Virtual machines", "Storage", and "Network". Under "Virtual machines", "VM instances" is selected. The main content area shows a table of VM instances. One instance is listed: "mygoogle-cloud1" in the "us-central1-c" zone. Below the table, there are "Related actions" such as "Fix data protection gaps", "View billing report", "Monitor VMs", "Explore VM logs", "Set up firewall rules", "Patch management", and "Load balance between VMs". On the right side, there's a "Get started with Compute Engine" tutorial section with links to various tutorials like "Create a 'hello world' website on IIS", "Create a 'hello world' website on Apache", "Transfer files to a Windows VM", "Transfer files to a Linux VM", "Enable ingress traffic", and "Back up and restore a VM".

Status	Name	Zone	Recommendation	Connect
☐	mygoogle-cloud1	us-central1-c		SSH

img1

XX

The screenshot shows the Google Cloud Platform console for the project 'liquid-engine-468502-s6'. The 'Compute Engine' section is active, and the 'VM instances' tab is selected. A table lists the VM instances, with 'mygoogle-cloud1' in the 'us-central1-c' zone. A context menu is open for this instance, showing various actions. The 'Start / Resume' action is highlighted.

Status	Name	Zone	Recommendation	Connect
☐	mygoogle-cloud1	us-central1-c		SSH

Related actions:

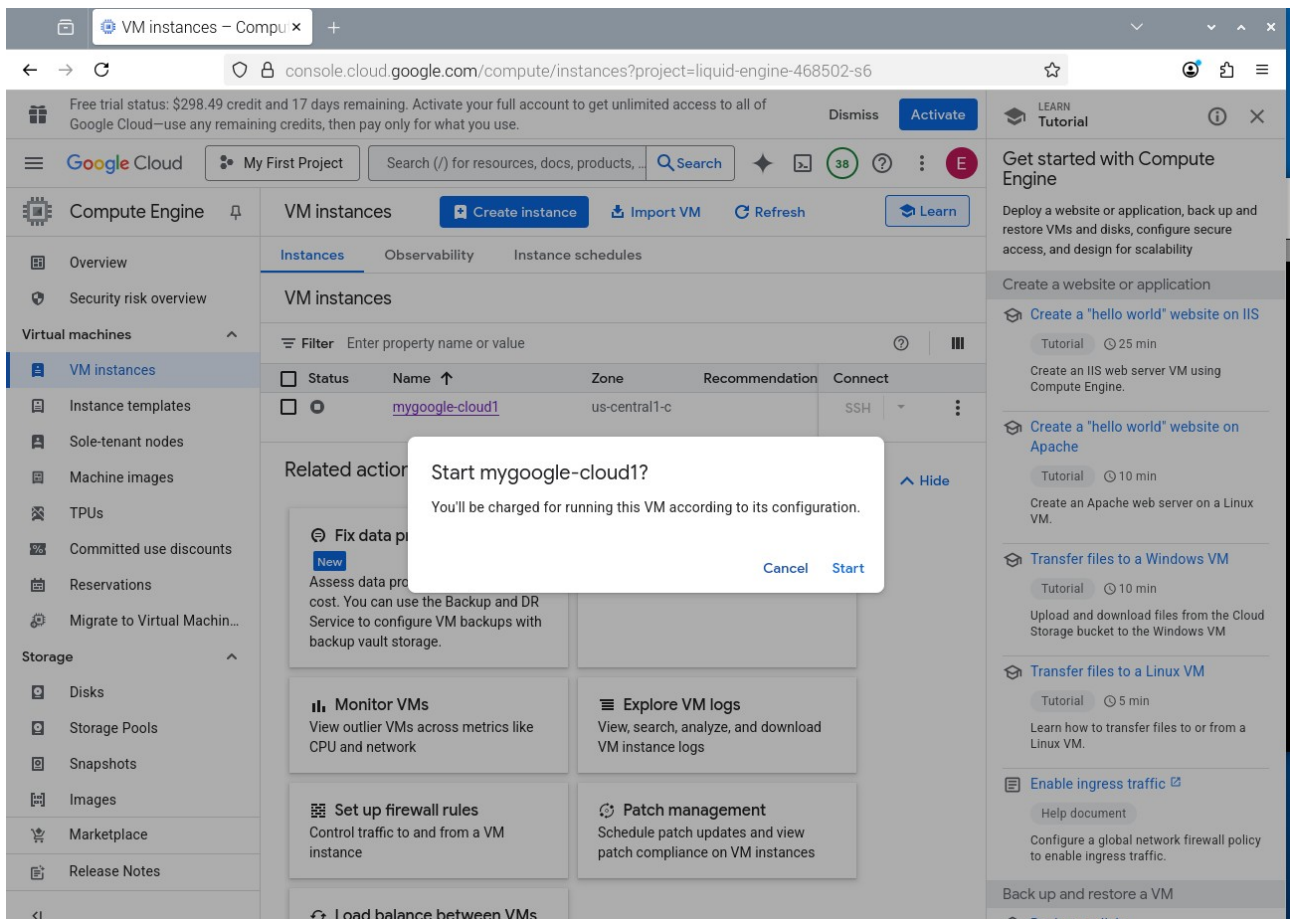
- Fix data protection gaps** (New): Assess data protection gaps at no cost. You can use the Backup and DR Service to configure VM backups with backup vault storage.
- View billing report**: View and manage your Compute Engine billing.
- Monitor VMs**: View outlier VMs across metrics like CPU and network.
- Explore VM logs**: View, search, analyze, and download VM instance logs.
- Set up firewall rules**: Control traffic to and from a VM instance.
- Patch management**: Schedule patch updates and view patch compliance on VM instances.
- Load balance between VMs**

Context menu actions:

- Start / Resume
- Stop
- Suspend
- Reset
- Run maintenance
- Configure data protection
- Delete
- Create a group based on this VM (Preview)
- View network details
- Create new machine image
- View logs
- View monitoring

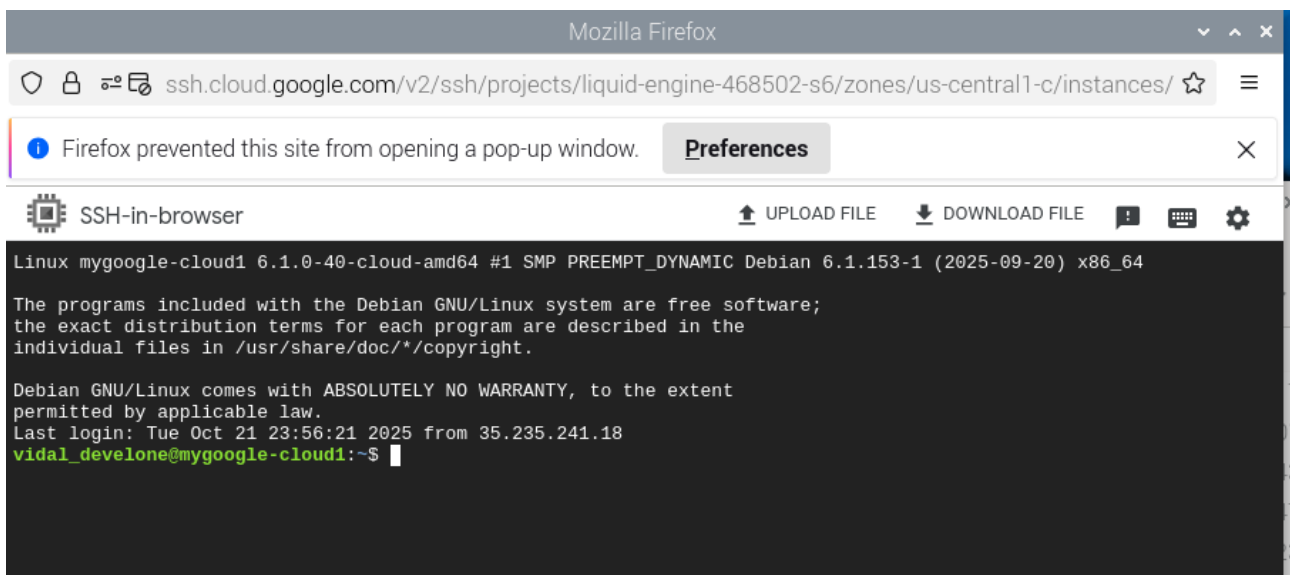
img2

After depressing Start/Resume The ssh is enable



Depress Start

img3



img4

```
Linux mygoogle-cloud1 6.1.0-40-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.153-1 (2025-09-20) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Tue Oct 21 23:56:21 2025 from 35.235.241.18
vidal_develone@mygoogle-cloud1:~$ . myenv.sh
/home/vidal_develone/oss-cad-suite/
/home/vidal_develone/PipelineC
vidal_develone@mygoogle-cloud1:~$
```

10/21/25 testing google cloud mygoogle-cloud1

```
PRETTY_NAME="Debian GNU/Linux 12 (bookworm)"
NAME="Debian GNU/Linux"
VERSION_ID="12"
VERSION="12 (bookworm)"
VERSION_CODENAME=bookworm
ID=debian
HOME_URL="https://www.debian.org/"
SUPPORT_URL="https://www.debian.org/support"
BUG_REPORT_URL="https://bugs.debian.org/"
```

```
vidal_develone@mygoogle-cloud1:~/pico-ice/PipelineC/blink-ex-uart-echo/
ice_makefile_pipelinec$ git branch -a
* test-dev-100424
```

```
vidal_develone@mygoogle-cloud1:~/PipelineC$ git branch -a
* test-dev
```

```
. myenv.sh
cat ~/myenv.sh
#!/bin/bash
export OSS_CAD_SUITE=$HOME/oss-cad-suite/
export PIPELINEC_REPO="$HOME/PipelineC"
echo $OSS_CAD_SUITE
echo $PIPELINEC_REPO
```

```
cd pico-ice/PipelineC/blink-ex-uart-echo/ice_makefile_pipelinec/
./build.sh
```

```
scp gateway.uf2 devel@100.89.157.49:~/mygoogle-cloud1/
```

```
cd ../
```

```
cd blink-ex-uart-echo/ice_makefile_pipelinec/
```


./build.sh

scp gateway.uf2 devel@100.89.157.49:~/mygoogle-cloud1/blink-ex-uart-echo/

devel@pi5-90:~ \$ minicom ACM1

ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789

img5

video

